COMP 3920

DATABASE SYSTEMS 1

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MORE TO A DATABASE

- Views
- Stored Procedures
- Stored Functions
- Triggers

VIEWS

- What is a View?

- Views are SELECT statements that get stored
- Once created they act like tables

WHY USE VIEWS?

- With Views we can:
  - Include/Exclude columns
  - Join multiple tables
  - Union multiple tables
  - Calculate column values (after_tax = price * (1+TAX_RATE))
  - Aggregate values (average, min/max, count, sum)
  - Can include tables or even other views
  - Enforce security

DISADVANTAGE OF VIEWS

- Possible Disadvantages of Views
  - Decreased performance
  - Changes to base tables need to be reflected in views (otherwise your view becomes inactive)
  - The data retrieved from views are often non-updatable

STORED PROCEDURE/FUNCTION

- What is a Stored Procedure / Function?
  - Sometimes called Routines
  - "A stored routine is a set of SQL statements that can be stored in the server." –MySQL Ref
  - Can also include code.
- **IMPORTANT NOTE:** `mysqldump` does NOT backup Stored Procedures / Functions (or routines) by default use the `--routines` option

https://dev.mysql.com/doc/refman/5.6/en/stored-routines.html

PROCEDURE VS FUNCTION

- Functions can be called as part of SELECT statements (Procedures cannot)
- Procedures are pre-compiled to increase performance
- Functions can not change the state of a transaction
- Some syntax differences
- A few other differences

https://dev.mysql.com/doc/refman/5.6/en/stored-routines.html

TRIGGERS

- What is a Trigger?

- A trigger is a set of code that is run when certain events happen on a table.
- Can call Stored Procedures.

TRIGGER EVENT TYPES

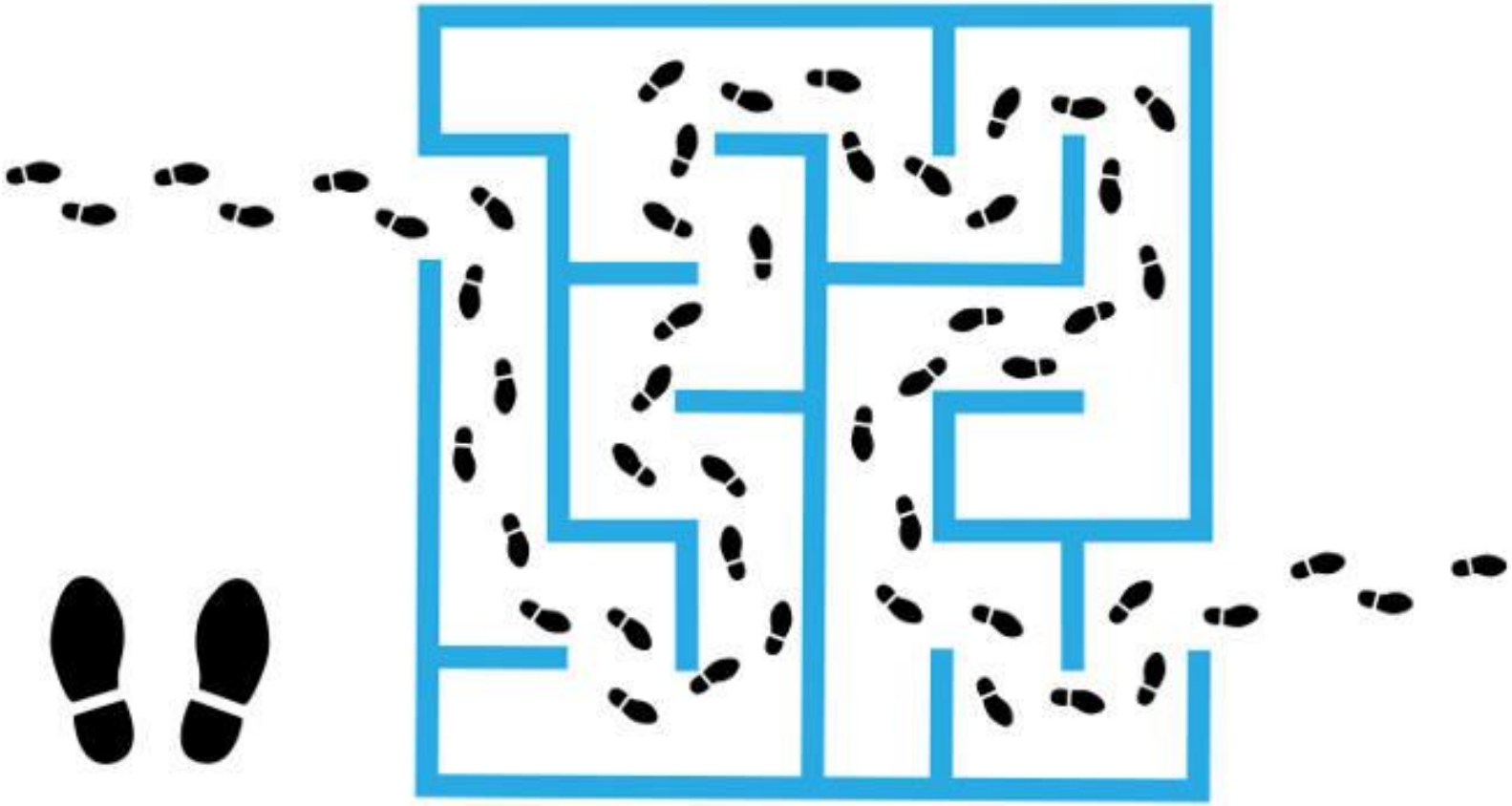
BEFORE

- INSERT
- UPDATE
- DELETE

AFTER

- INSERT
- UPDATE
- DELETE

LAB WALKTHROUGH



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